

Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study

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ABSTRACT

This paper contributes a benchmark and a leakage-controlled evaluation protocol for cross-year near-infrared (NIR) origin discrimination of commercial apples. The benchmark is a nine-domain dataset covering three harvest years (2018, 2019, 2025) and four origins (Xinjiang, Shandong, Shanxi, Gansu), with 2,424 fruits, 4,529 spectra and 229 bands over the 590–1001 nm overlap window. We define a fold-restricted leave-one-year-out (LOEO-Year) protocol with three components: (i) a unified single-face query rule applied to all methods on every fold; (ii) a strict source-only z -score pre-processing variant that is fully target-blind; and (iii) a shared-encoder versus fold-restricted-encoder split that quantifies evaluation leakage. A shared encoder inflated the three-fold macro F_1 at $K = 5$ by +0.266 (0.931 vs. 0.665) over the best fold-restricted configuration; the gap widened to +0.333 on the LOEO-2025 fold, which confounds year, measurement-protocol and class-space change. Fifty-repetition experiments over $K \in \{1, 3, 5, 10, 20, 50\}$ compared four few-shot baselines under the unified protocol—vanilla Prototypical Networks (ProtoNet), RBF-SVM, partial least squares discriminant analysis (PLS-DA) and nearest centroid—plus an exploratory ProtoNet-C source-centroid-fusion ablation that did not outperform vanilla ProtoNet on any fold. In the three-fold mean, vanilla ProtoNet leads for $K \leq 10$ and SVM overtakes for $K \geq 20$; the crossover is sensitive to source-domain composition. We also report 1D CNN and 1D ResNet supervised closed-set sanity checks. A 2×2 ablation on the 2025 fold decomposed the face-mean gain into a query-side contribution of +0.055 and a support-side contribution of +0.011 in F_1 , informing in-field scanning. Future LOEO studies should declare target-domain exposure and report a strict source-only baseline.

1. Introduction

The apple (*Malus domestica* Borkh.) is one of the most widely consumed temperate fruits in the world, and Chinese production accounts for more than 47% of the global supply. Geographic origin is an important attribute for both pricing and food-safety traceability, since price differentials between protected origins can be substantial and false labelling has well-documented economic and regulatory consequences. Conventional analytical approaches—stable-isotope-ratio analysis, multi-element fingerprinting and sensory evaluation—offer high specificity but require specialised laboratory infrastructure and detection cycles of several hours, which makes them unsuitable for batch-scale screening (Danezis, Tsagkaris, Camin, Brusci and Georgiou, 2016; Gonzalvez, Armenta and de la Guardia, 2009).

Near-infrared (NIR) diffuse-reflectance spectroscopy is an attractive alternative: the measurement is reagent-free, takes less than a minute per scan, and yields directly interpretable spectra

(Nicolai, Beullens, Bobelyn, Peirs, Saeys, Theron and Lammertyn, 2007; Blanco and Villarroya, 2002; Walsh, Blasco, Zude-Sasse and Sun, 2020). Spectral variation in apple fruit reflects the joint contribution of soluble solids, organic acids, polyphenols and tissue water content, all of which are influenced by soil geochemistry, climate and orchard management; the resulting high-dimensional patterns can provide an origin-related spectral fingerprint for apples under commercial conditions (Cen and He, 2007; Bureau, Cozzolino and Clark, 2019), in line with broader evidence that NIR can encode geographic fingerprints across food matrices (Woodcock, Downey, Kelly and O'Donnell, 2007). For commercial apples sourced from wholesale markets, NIR spectra additionally encode varietal differences, storage conditions, ripeness and supply-chain handling. Accordingly, we use “origin traceability” to mean origin-label discrimination under commercial-batch conditions. Partial least squares discriminant analysis (PLS-DA) is the long-established baseline for this task and typically achieves high accuracy under single-year, single-instrument conditions (Geladi and Kowalski, 1986; Wold, Sjöström and Eriksson, 2001; Westad and Marini, 2015). With larger training sets, support vector

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